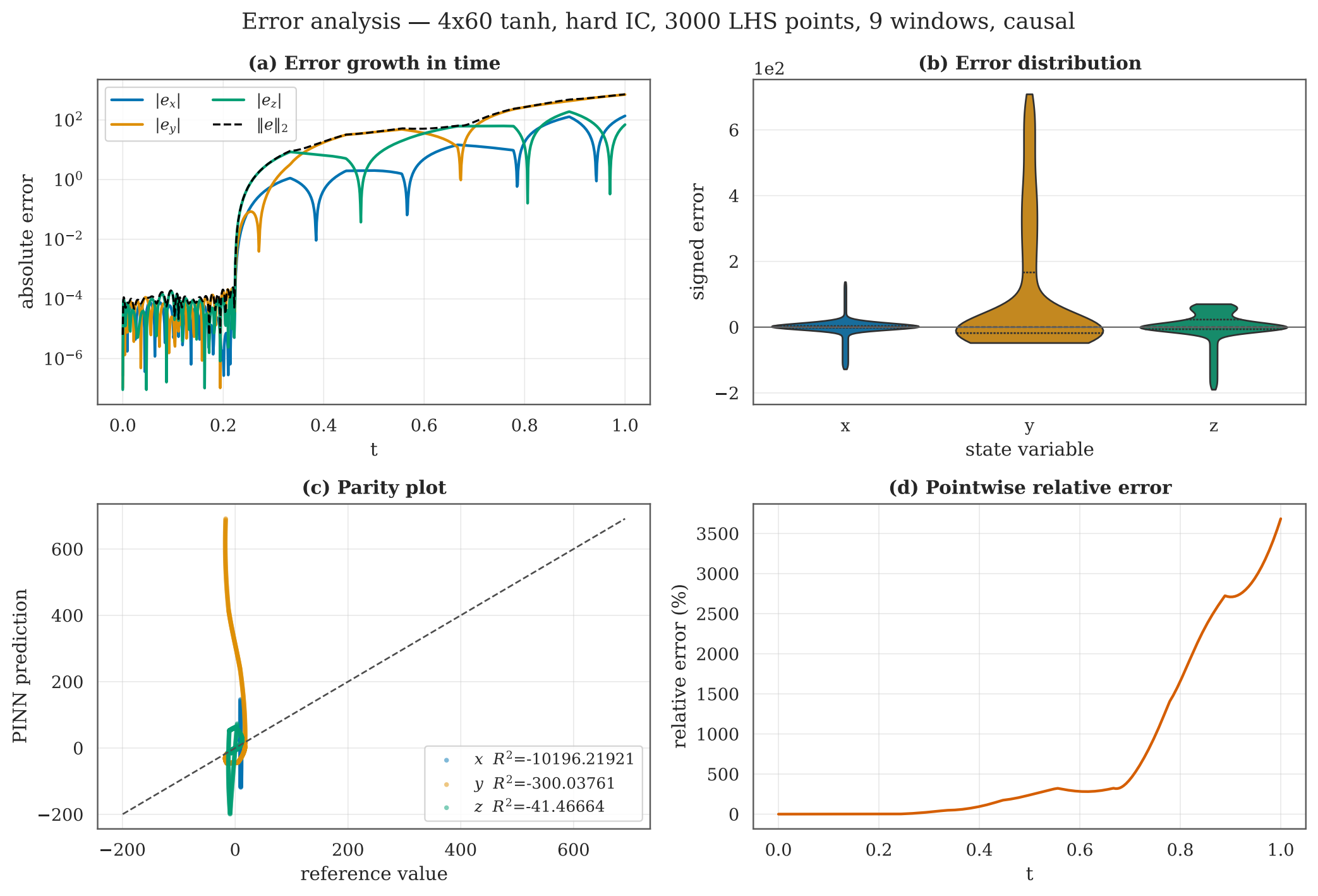

Error analysis — 4x60 tanh, hard IC, 3000 LHS points, 9 windows, causal
(a) Error growth in time
$|e_x|$
$|e_y|$
$|e_z|$
$\|e\|_2$
absolute error
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
0.0
0.2
0.4
0.6
0.8
1.0
t
(b) Error distribution
1e2
signed error
6
4
2
0
−2
x
y
z
state variable
(c) Parity plot
PINN prediction
600
400
200
0
−200
−200
0
200
400
600
reference value
x R^2=-10196.21921
y R^2=-300.03761
z R^2=-41.46664
(d) Pointwise relative error
relative error (%)
3500
3000
2500
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
t